In questa giornata del corso ho approfondito la fase di *Vulnerability Assessment (VA)* , ovvero la valutazione delle vulnerabilità presenti nei sistemi informatici.Questo rappresenta una tappa fondamentale nel processo di ethical hacking: permette di individuare configurazioni errate, servizi esposti o software non aggiornati, assegnando loro un livello di rischio. A differenza del Penetration Test , questa fase non include l'exploitation, ma prepara il terreno per eventuali exploit futuri.

L'obiettivo principale dell'esercitazione era effettuare una serie di scansioni con **Nmap** su un sistema **Windows** , sia con il **firewall attivo** che **disattivato** , al fine di confrontare i risultati ottenuti. Inoltre, è stato richiesto di documentare ogni passaggio, mostrare i comandi utilizzati e commentare le differenze osservate.

Configurazione della Rete

Prima di procedere con le scansioni vere e proprie, ho configurato due macchine virtuali su VMware:

- **Kali Linux** : utilizzato come sistema attaccante.
- **Windows 10/11** : il target da analizzare.

Ho assicurato che entrambe fossero sulla stessa rete virtuale e che potessero comunicare tra loro. Ho verificato la connettività tramite il comando `ping` e ho controllato gli indirizzi IP:

- **Kali Linux** : `192.168.222.129`
- **Windows** : `192.168.222.128`

Dopo aver confermato la comunicazione reciproca, sono passato alle scansioni con Nmap.

Scansioni con Firewall Attivo

Come prima cosa, ho lasciato il firewall di Windows abilitato e ho lanciato diverse tipologie di scansione con Nmap da Kali Linux.

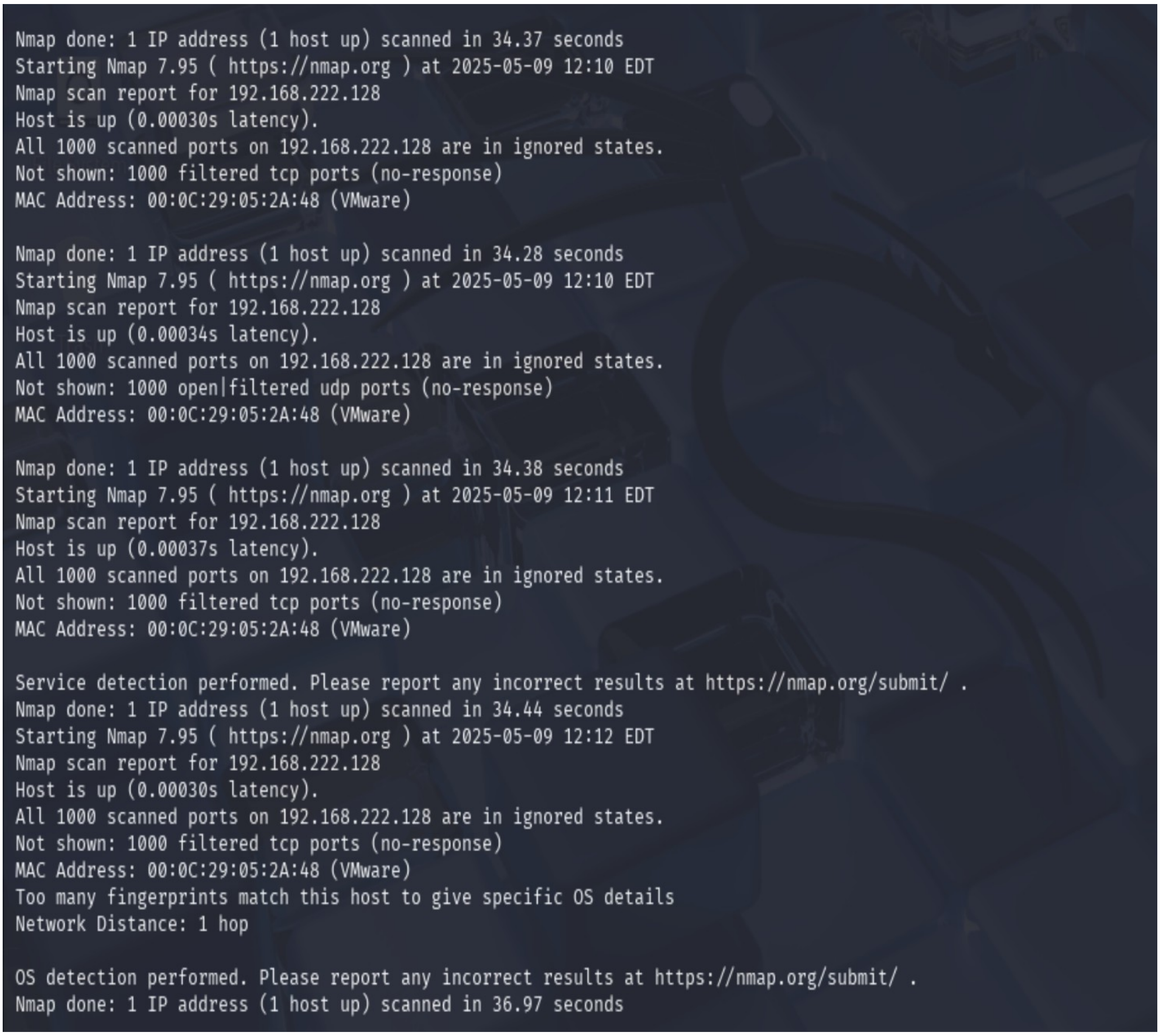
Considerazione : Con il firewall attivo, il sistema Windows appare molto più "invisibile" alla scansione esterna. Le porte non vengono mostrate, i servizi non sono riconoscibili e il sistema sembra quasi non esistere per l'attaccante.

```
nmap -sV 192.168.222.128
nmap -O 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:09 EDT
Nmap scan report for 192.168.222.1
Host is up (0.00023s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 192.168.222.128
Host is up (0.00012s latency).
MAC Address: 00:0C:29:05:2A:48 (VMware)
Nmap scan report for 192.168.222.254
Host is up (0.00019s latency).
MAC Address: 00:50:56:E6:3C:88 (VMware)
Nmap scan report for 192.168.222.129
Host is up.
Nmap scan report for 192.168.222.131
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 27.89 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:09 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00025s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 34.37 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:10 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00030s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 34.28 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:10 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00034s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 34.38 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:11 EDT
```



```
Nmap done: 1 IP address (1 host up) scanned in 34.37 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:10 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00030s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 34.28 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:10 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00034s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 open|filtered udp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 34.38 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:11 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00037s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.44 seconds
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:12 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00030s latency).
All 1000 scanned ports on 192.168.222.128 are in ignored states.
Not shown: 1000 filtered tcp ports (no-response)
MAC Address: 00:0C:29:05:2A:48 (VMware)
Too many fingerprints match this host to give specific OS details
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 36.97 seconds
```

Scansioni con Firewall Disattivato

A questo punto, ho disattivato temporaneamente il firewall di Windows, usando PowerShell con privilegi elevati:

Con il firewall disattivato, Nmap è riuscito a riconoscere chiaramente che si trattava di un sistema Windows, probabilmente tra la versione 7 e 10.

Considerazione : Senza il firewall, il sistema diventa molto più visibile e vulnerabile. I servizi esposti possono essere sfruttati, e la versione del sistema operativo è facilmente identificabile.

```
File  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ nmap -sn 192.168.222.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 11:42 EDT
Nmap scan report for 192.168.222.1
Host is up (0.00021s latency).
MAC Address: 00:50:56:C0:00:01 (VMware)
Nmap scan report for 192.168.222.128
Host is up (0.00026s latency).
MAC Address: 00:0C:29:05:2A:48 (VMware)
Nmap scan report for 192.168.222.254
Host is up (0.000085s latency).
MAC Address: 00:50:56:E6:3C:88 (VMware)
Nmap scan report for 192.168.222.129
Host is up.
Nmap scan report for 192.168.222.131
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 27.91 seconds

┌──(kali㉿kali)-[~]
└─$ ping 192.168.222.128
PING 192.168.222.128 (192.168.222.128) 56(84) bytes of data.
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sS 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 11:54 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00032s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 14.45 seconds
```

```
Nmap done: 1 IP address (1 host up) scanned in 16.17 seconds

┌──(kali㉿kali)-[~]
└─$ nmap -sU 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 11:55 EDT
Stats: 0:03:46 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Stats: 0:03:47 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Stats: 0:03:47 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Stats: 0:03:49 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Stats: 0:03:49 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Stats: 0:03:50 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 99.99% done; ETC: 11:59 (0:00:00 remaining)
Nmap scan report for 192.168.222.128
Host is up (0.00032s latency).
Not shown: 907 closed udp ports (port-unreach), 92 open|filtered udp ports (no-response)
PORT    STATE SERVICE
137/udp open  netbios-ns
MAC Address: 00:0C:29:05:2A:48 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 237.80 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:00 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00019s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 00:0C:29:05:2A:48 (VMware)
Service Info: Host: WIN-UVO9VJTFD3R; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 73.36 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -sV 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:00 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00019s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
139/tcp   open  netbios-ssn  Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds Microsoft Windows 7 - 10 microsoft-ds (workgroup: WORKGROUP)
49152/tcp open  msrpc        Microsoft Windows RPC
49153/tcp open  msrpc        Microsoft Windows RPC
49154/tcp open  msrpc        Microsoft Windows RPC
49155/tcp open  msrpc        Microsoft Windows RPC
49156/tcp open  msrpc        Microsoft Windows RPC
49157/tcp open  msrpc        Microsoft Windows RPC
MAC Address: 00:0C:29:05:2A:48 (VMware)
Service Info: Host: WIN-UVO9VJTFD3R; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 73.36 seconds
```

```
┌──(kali㉿kali)-[~]
└─$ nmap -O 192.168.222.128
Starting Nmap 7.95 ( https://nmap.org ) at 2025-05-09 12:04 EDT
Nmap scan report for 192.168.222.128
Host is up (0.00035s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49155/tcp open  unknown
49156/tcp open  unknown
49157/tcp open  unknown
MAC Address: 00:0C:29:05:2A:48 (VMware)
Device type: general purpose
Running: Microsoft Windows 2008|7|Vista|8.1
OS CPE: cpe:/o:microsoft:windows_server_2008:r2 cpe:/o:microsoft:windows_7 cpe:/o:microsoft:windows_vista cpe:/o:microsoft:windows_8.1
OS details: Microsoft Windows Server 2008 R2 SP1 or Windows 7 SP1, Microsoft Windows Vista SP2 or Windows 7 or Windows Server 2008 R2 or Windows 8.1
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 19.68 seconds
```

Confronto Risultati: Firewall ON vs OFF

Tipo di Scansione	Risultato con FW On	Risultato con FW Off
`-sn`(Ping Sweep)	Host rilevato	Host rilevato
`-sS`(SYN Scan)	Nessuna porta aperta	Porte aperte individuate
`-sT`(TCP Connect)	Nessuna porta aperta	Porte aperte individuate
`-sU`(UDP Scan)	Nessuna porta UDP visibile	Individuata porta`137/udp`(NetBIOS)
`-sV`(Service Detect)	Servizi non riconosciuti	Servizi e versioni ben identificate
`-O`(OS Detection)	OS non riconosciuto	Sistema riconosciuto come Windows

Osservazione finale: La presenza del firewall ha ridotto drasticamente la superficie d'attacco visibile dall'esterno. Molti servizi non sono stati rilevati, e nemmeno la versione del sistema operativo è stata facile da individuare. Tuttavia, con il firewall disattivato, il sistema è diventato molto più debole e facilmente attaccabile.

Conclusioni

Questo esercizio mi ha aiutato a comprendere quanto sia importante il firewall in un sistema operativo, soprattutto per proteggere porte sensibili come quelle dei servizi SMB (`445/tcp`) o RPC (`135/tcp`). Lavorando con Nmap, ho imparato a riconoscere le principali tipologie di scansione e a interpretarne i risultati. Ho anche visto come cambia radicalmente il comportamento di un sistema a seconda dello stato del firewall.

Sebbene il firewall non blocchi completamente un attaccante avanzato, riesce comunque a rendere il lavoro molto più difficile, nascondendo informazioni preziose come il sistema operativo e i servizi esposti.